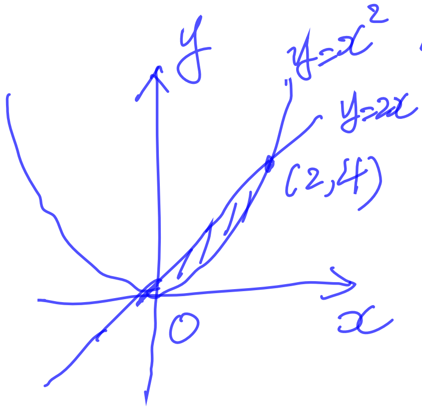
Math 142, F20 **Name:** Answer

Quiz 2

Instructions: **You must show all of your work to receive credits for correct answers.**

Problem 1. (4 points) Find the area of the region bounded by the curves $y = 2x$ and $y = x^2$.



$\begin{cases} y = 2x \\ y = x^2 \end{cases} \Rightarrow x^2 = 2x \Rightarrow x^2 - 2x = 0$

$x(x-2) = 0$

$\Rightarrow x = 0, \; x = 2$

$(y = 0), \; (y = 4)$

$\text{Area} = \int_0^2 2x - x^2 \, dx$

$= \left[x^2 - \frac{x^3}{3}\right]_{x=0}^{x=2} = \left[2^2 - \frac{2^3}{3}\right] - \left[0^2 - \frac{0^3}{3}\right]$

$= 4 - \frac{8}{3} = \frac{4}{3}$

Problem 3. (6 points) Evaluate the following integrals:

(a) $\displaystyle\int \frac{1}{e^{-x} + e^{x}} \, dx$

$\int \frac{1}{e^{-x}+e^{x}} dx = \int \frac{1 \cdot e^{x}}{(e^{-x}+e^{x}) \cdot e^{x}} dx$

$= \int \frac{e^{x}}{1+e^{2x}} dx \xrightarrow[du = e^{x}dx]{u = e^{x}} \int \frac{du}{1+u^2}$

$= \arctan u + C$

$= \arctan(e^{x}) + C$

(b) $\displaystyle\int \sqrt{x} \ln x \, dx$

let $u = \ln x$, $dv = \sqrt{x}\, dx$

$du = \frac{1}{x} dx$, $v = \frac{2}{3} x^{\frac{3}{2}}$

$\Rightarrow \int \sqrt{x} \ln x \, dx$

$= \frac{2}{3} x^{\frac{3}{2}} \ln x - \int \frac{2}{3} x^{\frac{3}{2}} \cdot \frac{1}{x} dx$

$= \frac{2}{3} x^{\frac{3}{2}} \ln x - \frac{2}{3} \int x^{\frac{1}{2}} dx$

$= \frac{2}{3} x^{\frac{3}{2}} \ln x - \frac{4}{9} x^{\frac{3}{2}} + C$

(c) $\displaystyle\int_0^1 t e^{-2t} \, dt$

let $u = t$, $dv = e^{-2t} dt$

$du = dt$, $v = -\frac{1}{2} e^{-2t}$

$\Rightarrow \int_0^1 t e^{-2t} dt = t \cdot \left(-\frac{1}{2} e^{-2t}\right)\Big|_{t=0}^{t=1}$

$- \int_0^1 \left(-\frac{1}{2} e^{-2t}\right) dt$

$= -\frac{1}{2} e^{-2} - \left[\frac{1}{4} e^{-2t}\right]_{t=0}^{t=1}$

$= -\frac{1}{2} e^{-2} - \frac{1}{4} e^{-2} + \frac{1}{4} = -\frac{3}{4} e^{-2} + \frac{1}{4}$

(d) $\displaystyle\int \sin^{-1} x \, dx$

let $u = \sin^{-1} x$, $dv = dx$

$du = \frac{1}{\sqrt{1-x^2}} dx$, $v = x$

$\Rightarrow \int \sin^{-1} x \, dx = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx$

$\int \frac{x}{\sqrt{1-x^2}} dx \xrightarrow[du = -2x\,dx]{u = 1-x^2} \int -\frac{1}{2} \frac{1}{\sqrt{u}} du$

$= -\sqrt{u} + C = -\sqrt{1-x^2} + C$

thus $\int \sin^{-1} x \, dx = x \sin^{-1} x + \sqrt{1-x^2} + C$